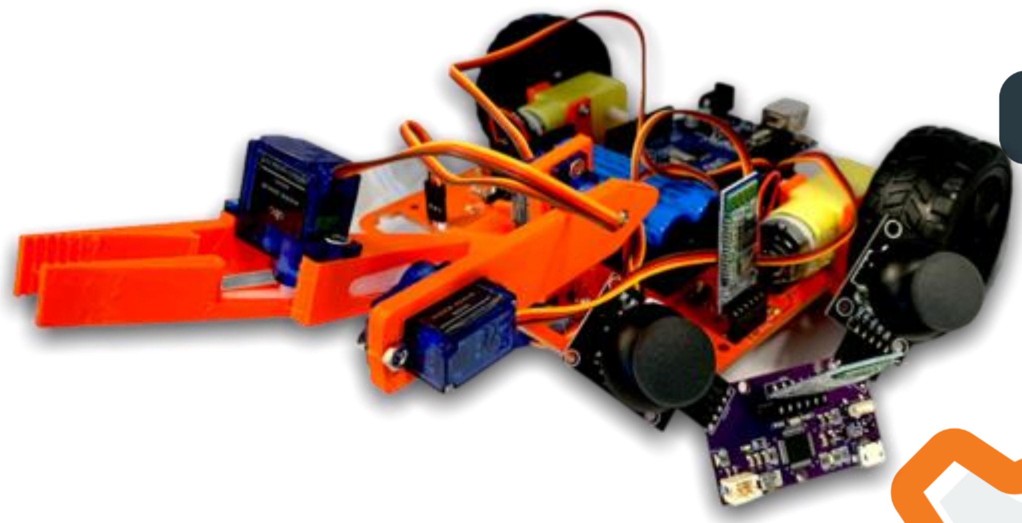
JUNIOR ROBOTICS

MANUAL

Emergency Bot



AEROBOTICS GLOBAL
www.aeroboticsglobal.com
info@aeroboticsglobal.com
1800 237 627

CONTENTS

WARNING: The kit contents are not suitable for children under 3 years. Choking hazard – small parts may be swallowed or inhaled.

Store the kit materials and assembled models out of reach of small children.

HANDLING AND PRECAUTIONS

General:

- Assembly and operation of this kit should be done only under the supervision of an adult
- All parts of this kit should be stored and operated away from water and fire sources
- All parts of this kit should never be put into the mouth
- Throwing or dropping the parts may cause damage to the parts

Building/Operating:

- All the tools for assembling must be used with extreme precautior
- The parts should not be forcibly bent, attached or removed
- The parts should not be touched when in motion
- The power supply, motors, sensors and other components should before operating
- The power supply should be turned off when connecting the Arduino to the computer for programming

Batteries:

- The batteries should not be inserted in the battery holder before mounting and connecting the battery holder onto the model
- The batteries should be inserted correctly with the (+) and (-) in the correct direction
- Always use the provided batteries or batteries of correct size and rating (3.7V)
- New and used batteries should not be mixed for operation
- Always close the battery holder with the lid after removal or insertion of batteries

Workshop 1 & 2

Hi Andy, welcome to the next part of our course which is to make an Emergency bot

What is an Emergency bot?

An Emergency Bot is a robot which can be used in dangerous situations to pick up and throw any objects like bombs or landmines etc..

Interesting, I'm excited to build one

Great!!! Today we are learning how to build an emergency bot with an arm and start assembling the robotic arm with servo motors

Now, before assembling, the main three key parts of the robotic arm are Base, Extender and the Gripper as shown in the figures 1.01, 1.02 and 1.03

Figure 1.01

Figure 1.02

Figure 1.03

Building the robotic arm is a complex process, which we will execute in a step by step method by assembling gripper first, then extender and finally the base as follows

Step 1

Take a servo motor and place a semi horn on top of it as shown in figure 1.04, mount it on to the gripper as shown figure 1.05 and fit the servo with the screws as shown in figure 1.06

Figure 1.04

Figure 1.05

Figure 1.06

Step 2

After fitting the servo motor, place the 2nd half of the gripper on the semi horn at bottom end and tighten it with the screw as shown in figure 1.08

Figure 1.07

Figure 1.08

Figure 1.09

Step 3

Now take the extender and assemble a servo motor as shown in figure 1.10 and place a horn with only two long legs as shown in figure 1.11

Figure 1.10

Figure 1.11

Step 4

Now mount the gripper on top of the extender and tighten it with screw as shown
In figure 1.12

Figure 1.12

Figure 1.13

Step 5

Now take the base and insert and assemble another servo motor on top
of the base as shown in figure 1.14

Figure 1.14

Figure 1.15

Step 6

Now insert a full horn with four legs into the extender and mount the Base on top of it and tighten it with a screw as shown in figure 1.17

Figure 1.16

Figure 1.17

We now have successfully completed assembling the robotic arm as shown in figure 1.18, make sure your arm should look similar, if you have made any error, refer to earlier steps and reassemble

Figure 1.18

Workshop 3&4

Hi Aebo, what are we doing today?

We are learning about how to connect the robotic arm to Aebo and program it

What does this robotic arm do?

Robotic arm is the main part where we code the robot to performs the actions

Sounds good, Let's start

Insert a servo motor and assemble it with screws and place a four leg horn
On top of it as shown in figure 1.15

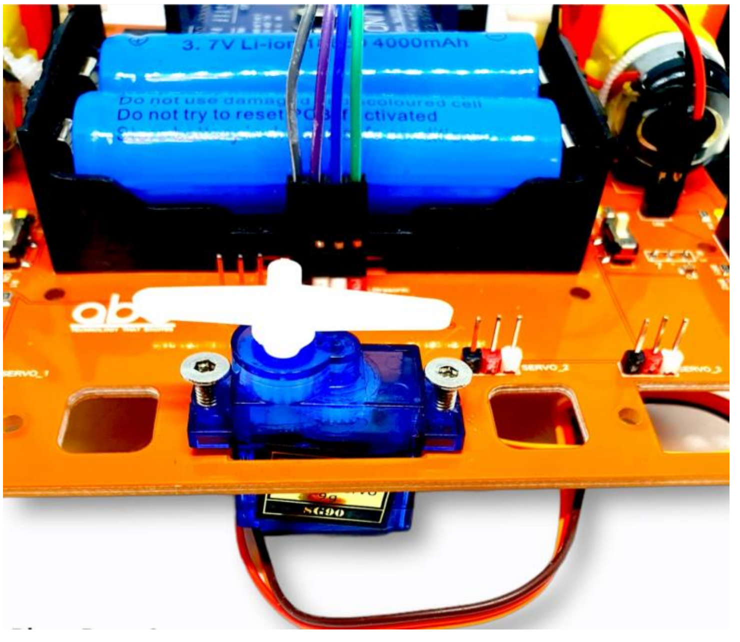


Figure 1.19

Connections

Connect the gripper servo motor to port 8 as shown in figure 1.21

Figure 1.20

Figure 1.21

Connect the extender servo motor to port 6 as shown in figure 1.23

Figure 1.22

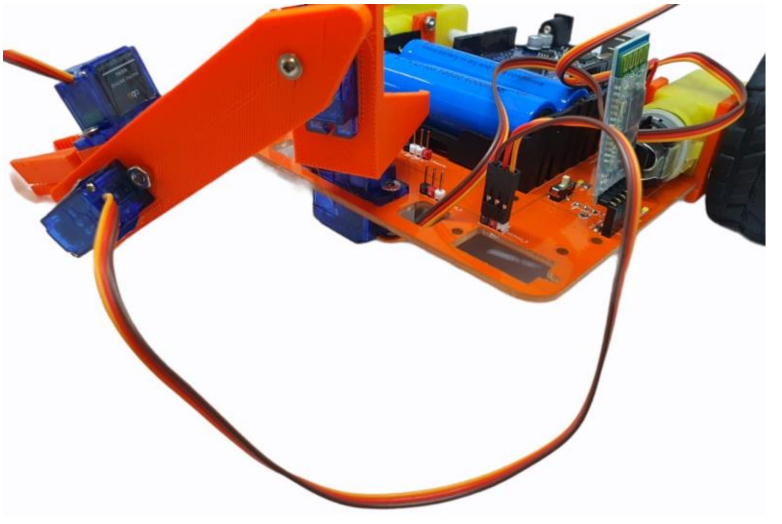


Figure 1.23

Connect the base servo motor to port 9 and connect the servo motor attached to the board into port 10 as shown in the figure 1.25

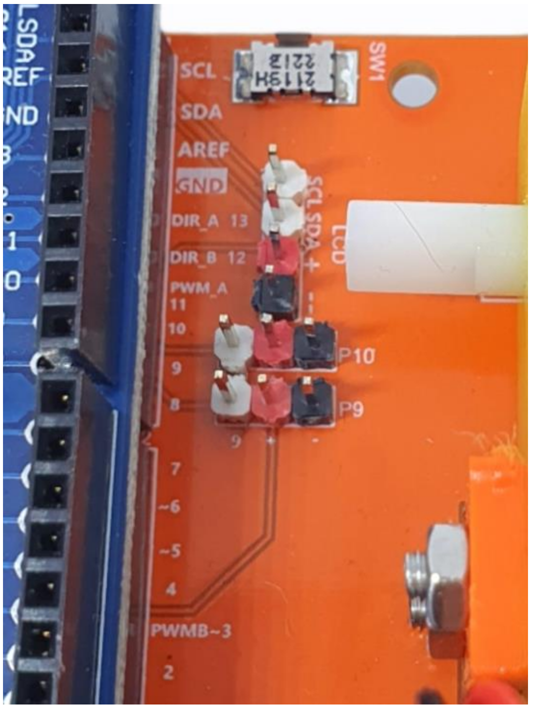


Figure 1.24

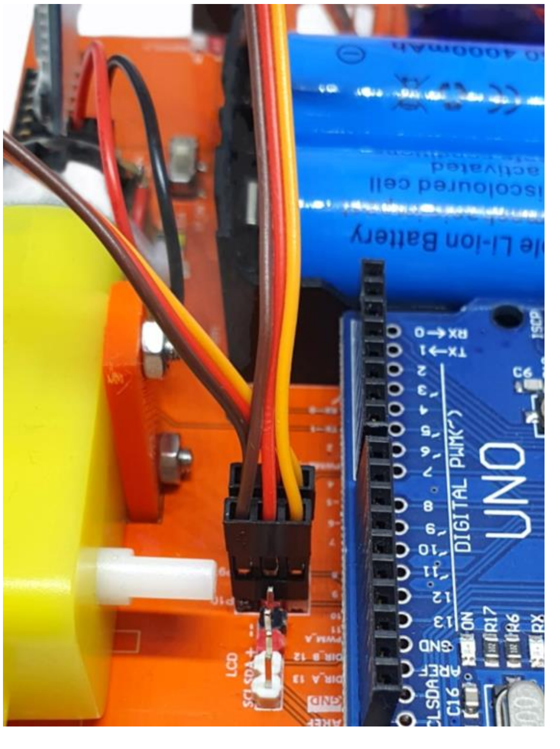


Figure 1.25

Now before mounting the robotic arm, first upload the calibration program in figure 1.26
To the Arduino which will set the arm in a straight line

Figure 1.26

After uploading the code, then adjust the servo horns accordingly so that
The robotic arm should be straight and then mount the arm on the servo motor
As shown in figure 1.28

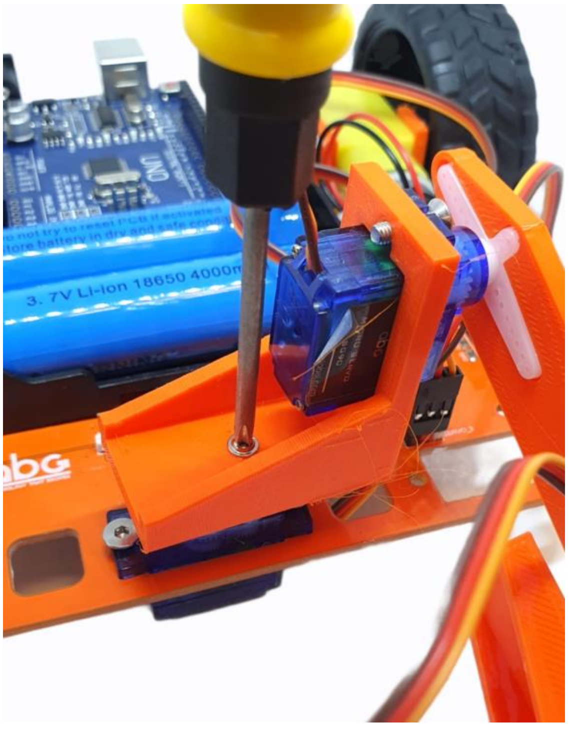
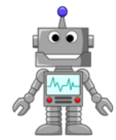


Figure 1.27

Figure 1.28

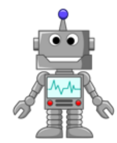
Workshop 5&6

Hi Aebo, what are we doing today?



We are coding our joystick and revising about connecting Bluetooth

Yes, I remember how we connected the bluetooths and coded the Joystick in Level 3 course



Then, Let’s start

Remember in Level 3 how we have used the joystick 2 to control forward, backward, left right and Stop with switch button, adding to that we use another joystick with the four positions of the gear can be used to control the Base servo(connected to port 10) movement Left and Right, and to move the robotic arm(connected to ports 6 & 9) in Up and Down directions. Initially when the gear is in idle position the X and Y values are 512,512 which means in an idle state and when the button is pressed once the arm will pick up the object and when pressed again it will release the object.

Now we have to make sure that the robotic arm should always be parallel to the ground, so that the servo motors in the base and extender should be moved together so that the arm stays parallel even in motion

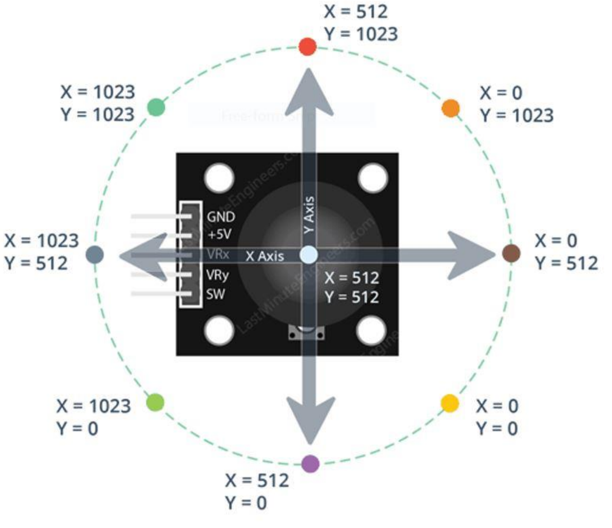


Figure 1.29

Now when both the Bluetooth modules were turned on, the red light will blink continuously which means it is searching for its pair, if we bring both of them within range then they will communicate and get paired, after that both Bluetooth lights will blink at heartbeat rate, insert the Bluetooth receiver module which has R written on its back side into the robot

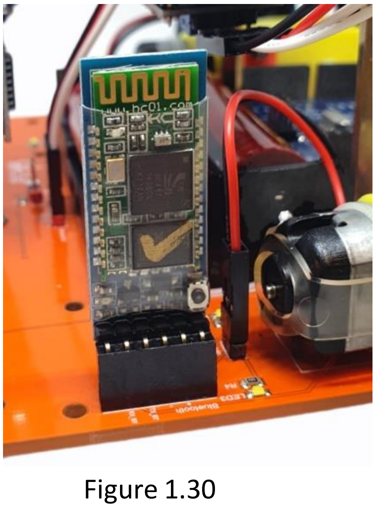


Figure 1.30

Figure 1.31

This code was already uploaded to control the Joystick

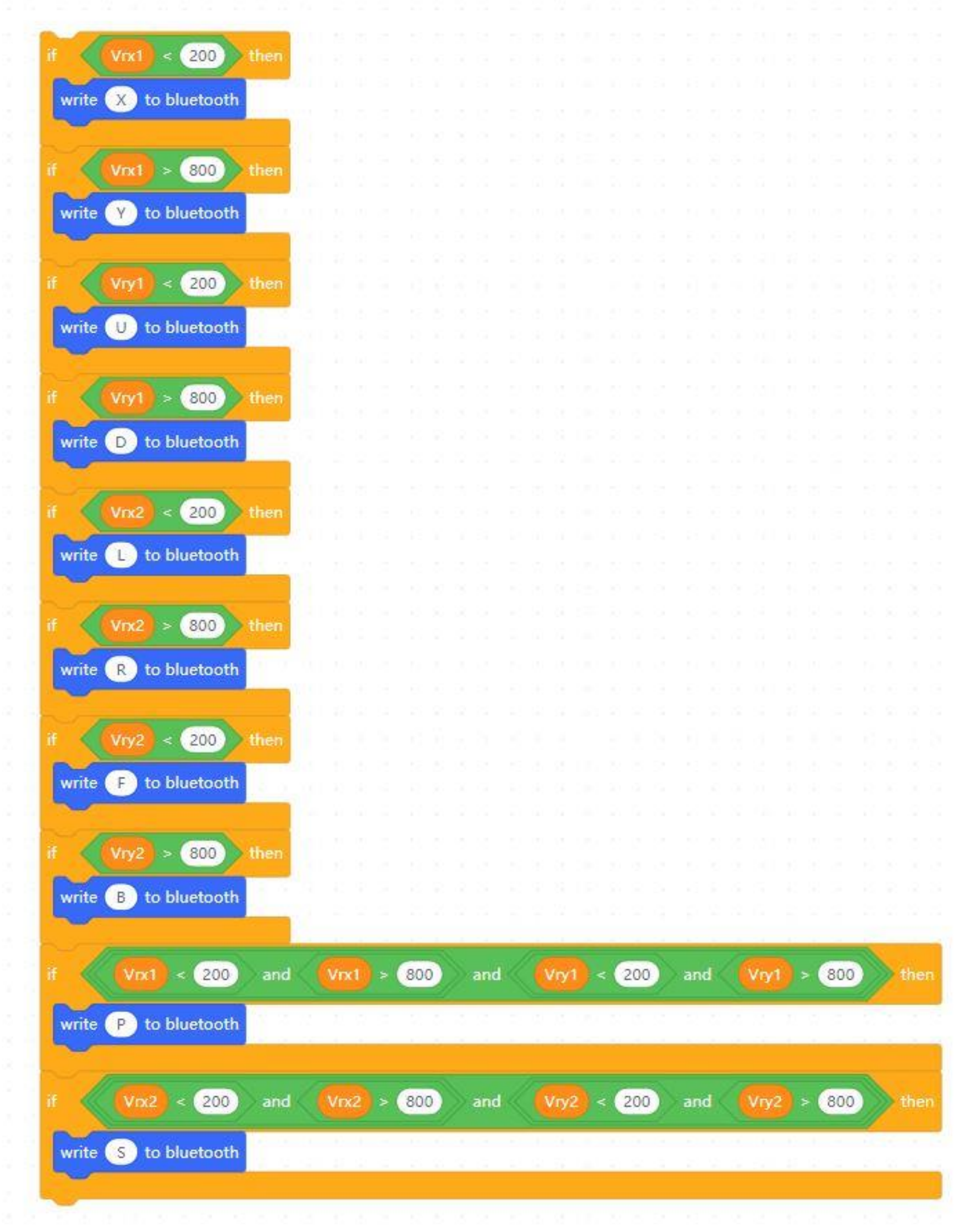
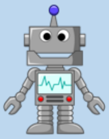


Figure 1.32

Workshop 7 & 8

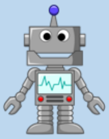
Hi Aebo, what are we doing today?

Today, we are defining blocks and finish building the emergency bot code



Yeah! Finally we are here to complete the course

Yes that’s right, Let’s start



We can use earlier defined blocks of Forward, Backward, Left Right and Stop_robot for one Joystick and define another set of blocks for second Joystick as shown below, we are shifting the robotic arm position by 3 degrees which gives a clear idea how it is moving, you can increase the change in position to a faster pace by moving 10 degrees as well

Figure 1.33

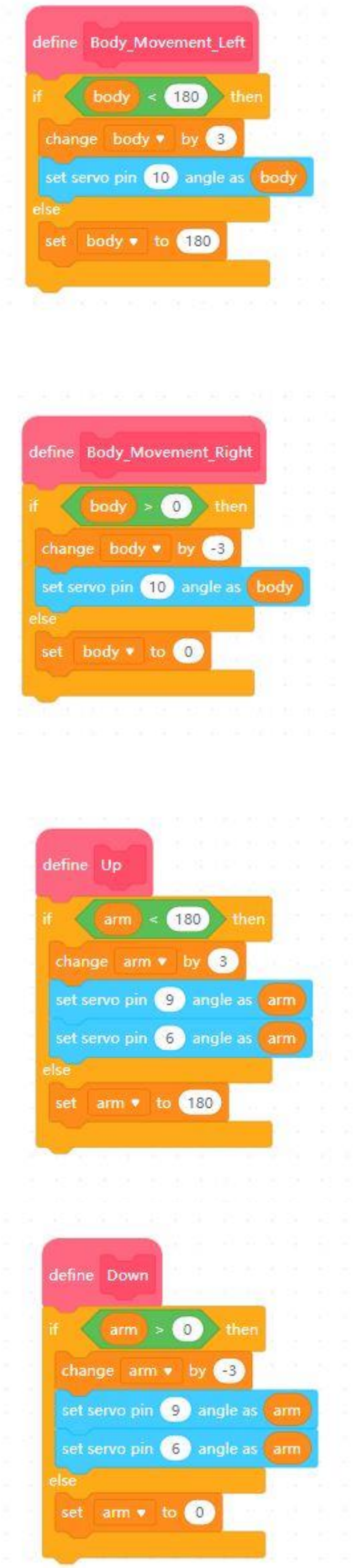


Figure 1.34

Final code for completing the Emergency Bot

Figure 1.35